

Polygraph System – Border Guard Officers

Device: Standard Polygraph System

Audience: Certified Border Guard Personnel

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Document Type: Operating Instructions

Classification: Internal Use – Border Guard Officers

1. Purpose and Scope

This document provides standardized operating instructions for the safe and effective use of the polygraph system during investigative interviews and candidate screening procedures.

The polygraph is used to:

- Support structured interview processes
- Assess credibility during recruitment
- Corroborate or challenge statements
- Identify physiological indicators of deception

NOTE: The polygraph is an investigative aid. Results must not be interpreted as definitive proof of truthfulness or deception.

2. General Safety Requirements

WARNING – LEGAL AND ETHICAL COMPLIANCE: Polygraph examinations must only be conducted by certified personnel and with informed consent.

- Participation must be voluntary.
- The subject must be informed of the purpose, procedure, and data handling.
- The examiner must follow internal Border Guard ethical guidelines.

CAUTION – EQUIPMENT HANDLING: Incorrect sensor placement or calibration may invalidate results.

- Do not overtighten pneumograph tubes or cuffs.
- Ensure all cables are free from tension and tripping hazards.
- Keep liquids away from the device.

NOTE: Subjects with certain medical conditions (cardiac disorders, respiratory issues, neurological conditions) may require medical clearance.

3. System Overview

The polygraph system records involuntary physiological responses to structured questions.

The system includes:

Component	Function
Pneumograph Tubes	Measure chest and abdominal respiration
Blood Pressure Cuff	Measures cardiovascular activity
Galvanic Skin Sensors	Measure electrodermal activity (EDA)
Control Unit	Collects and processes physiological
Software Interface	Displays real-time charts and stores test data

NOTE: Male subjects typically show stronger abdominal respiration; female subjects often show stronger thoracic respiration. Both must be monitored.

4. Physiological Signals Measured

Sensor Type	Measures	Possible Indicator of Deception
Pneumograph Tubes	Respiration rate and depth	Breath-holding, irregular breathing
Blood Pressure Cuff	Heart rate and blood pressure	Sudden spikes or instability
Galvanic Skin Response	Skin conductivity	Increased conductivity due to stress

NOTE: Physiological reactions may also result from anxiety, fatigue, or misunderstanding of questions.

5. Pre-Operation Procedure

5.1 Preparing the Subject

1. Explain the purpose and voluntary nature of the test.
2. Obtain written informed consent.
3. Ensure the subject is seated comfortably, feet flat, hands relaxed.
4. Ask the subject to minimize movement during the test.

WARNING: Do not proceed without explicit consent.

5.2 Attaching Sensors

1. Place pneumograph tubes around the chest and abdomen.
2. Wrap the blood pressure cuff around the upper arm.
3. Attach galvanic skin sensors to the index and ring fingers.
4. Ensure all sensors are snug but not restrictive.

CAUTION: Excessive pressure may cause discomfort or distort readings.

5.3 Software Startup and Calibration

1. Power on the polygraph control unit.
2. Launch the polygraph software.
3. Verify sensor connections.
4. Calibrate baseline readings for respiration, cardiovascular activity, and skin conductivity.
5. Confirm signal quality before proceeding.

NOTE: Calibration must be repeated if the subject adjusts posture.

6. Test Procedure – Border Guard Screening

6.1 Pre-Test Interview

Conducted by a certified examiner to:

- Explain the procedure and reduce anxiety
- Establish rapport
- Review the subject's background
- Clarify inconsistencies
- Confirm voluntary participation

WARNING: The examiner must not use coercive language or imply punitive consequences.

6.2 Questioning Protocol

Questions are categorized as follows:

Type	Purpose	Example
Relevant	Directly related to integrity or application	"Have you ever provided false information on your application?"
Control (Comparison)	Establish physiological baseline	"Have you ever lied to avoid getting into trouble?"
Irrelevant	Stabilize physiological patterns	"Is today Wednesday?"

NOTE: All questions must be phrased neutrally and delivered with consistent tone and timing.

6.3 Sample Question Sequence

1. "Is your name [Full Name]?" (Irrelevant)
2. "Have you ever used illegal drugs?" (Relevant)
3. "Have you ever lied to a supervisor?" (Control)
4. "Is your age 34?" (Irrelevant/Control)
5. "Have you ever committed a criminal offense not disclosed in your application?" (Relevant)
6. "Have you ever stolen anything of value?" (Control)
7. "Are you currently employed?" (Irrelevant)

CAUTION: Movement, coughing, or repositioning may invalidate the chart.

7. Monitoring and Interpretation

During the test, the examiner monitors:

- Respiration patterns
- Cardiovascular activity
- Galvanic skin response

Indicators of possible deception include:

- Sudden respiration suppression
- Cardiovascular spikes
- Increased skin conductivity

NOTE: Interpretation must consider context, subject anxiety, and baseline variability.

8. Post-Test Procedure

1. Review chart recordings for anomalies.
2. Identify questions associated with significant physiological reactions.
3. Discuss results with the subject (optional, depending on protocol).
4. Save and securely store test data.
5. Document findings in the internal system.

WARNING: Polygraph results must not be disclosed outside authorized personnel.

9. Ethical and Operational Requirements

- Ensure voluntary participation.
- Maintain subject dignity and comfort.
- Avoid leading or accusatory questions.
- Use results as part of a broader investigative strategy.
- Follow internal data retention and confidentiality policies.

NOTE: The polygraph is a support tool, not a standalone decision-making instrument.

10. Maintenance and Equipment Care

- Clean sensors after each use with approved disinfectant.
- Inspect cables and connectors weekly.
- Update software as released.
- Store equipment in a dry, temperature-controlled environment.
- Replace worn straps, cuffs, or sensors as needed.

CAUTION: Do not expose sensors to moisture or extreme temperatures.

11. Applicable Documents

- ISO 20607 – Safety of Machinery: Instruction Handbook Principles
- ISO 12100 – Risk Assessment and Risk Reduction
- ISO 10075 – Ergonomic Principles Related to Mental Workload
- ISO 9241-210 – Human-System Interaction
- Polygraph Manufacturer's Technical Manual
- Internal Data Protection and Retention Policy